REACHING A POINT OF THE SCHEME THEORETIC IMAGE BY A VALUATION RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a descent to the scheme case through an affine etale cover; every step of the descent is written out, including the base change identification of the scheme theoretic image.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Scheme theoretic image computed after base change by an etale morphism. Lemma lemma-quasi-compact-scheme-theoretic-image, used at proof lines 3326-3328. Identifies Z' with Z x_Y V so that z' lies in Z'.
- The scheme case of reaching points of a scheme theoretic image. Morphisms, Lemma morphisms-lemma-reach-points-scheme-theoretic-image, used at proof lines 3334-3348. Is applied to the morphism W -> V of schemes.
- Affine etale covers of quasi-compact algebraic spaces. used at proof lines 3329-3333. Produces the affine scheme W mapping surjectively etale onto X x_Y V.
- Affine schemes and etale morphisms hitting a prescribed point. used at proof lines 3324-3325. Produces V with the point z' mapping to z.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02JQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3345–3345. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact morphism of algebraic spaces over S. Let Z be the scheme theoretic image of f. Let $z \in |Z|$.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

There exists a valuation ring A with fraction field K and a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & & \longrightarrow & & X \\ \downarrow & & & \swarrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Z & \longrightarrow & Y \end{array}$$

such that the closed point of $\mathrm{Spec}(A)$ *maps to* z.

Proof. Choose an affine scheme V with a point $z' \in V$ and an étale morphism $V \to Y$ mapping z' to z. Let $Z' \subset V$ be the scheme theoretic image of $X \times_Y V \to V$. By Lemma 082Z we have $Z' = Z \times_Y V$. Thus $z' \in Z'$. Since f is quasi-compact and V is affine we see that $X \times_Y V$ is quasi-compact. Hence there exists an affine scheme W and a surjective étale morphism $W \to X \times_Y V$. Then $Z' \subset V$ is also the scheme theoretic image of $W \to V$. By Morphisms, Lemma 02JQ we can choose a diagram

$$\begin{array}{ccccccc} \mathrm{Spec}(K) & \longrightarrow & W & \longrightarrow & X \times_Y V & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Z' & \longrightarrow & V & \longrightarrow & Y \end{array}$$

such that the closed point of $\mathrm{Spec}(A)$ maps to z'. Composing with $Z' \to Z$ and $W \to X \times_Y V \to X$ we obtain a solution. □